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3.2.S.4.2 Analytical Procedures

The Analytical procedures for the tests described under the section **3.2.S.4.1 *Specification*** are appended in following pages.

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B. TEST PROCEDURE:

S.No.	Test Procedure
1.0	Description: By visual observation. (Spread sufficient amount of sample in a clean and dry petri dish / watch glass and visually observe the sample. Record the observation).
2.0	Solubility: <u>Soluble in water:</u> Weigh and transfer about 1.0 g of sample into a clean and dry stoppered test tube. Add 30 mL of water. Close the test tube with stopper and shake well manually. The resulting solution should be clear without any suspended particles. <u>Very slightly soluble in anhydrous ethanol:</u> Weigh and transfer about 10.0 mg of sample into a clean and dry volumetric flask. Add 100 mL of anhydrous ethanol. Close the test tube with stopper and shake well manually. The resulting solution should be clear without any suspended particles. <u>Practically insoluble in heptane:</u> Weigh and transfer about 10.0 mg of sample into a clean and dry volumetric flask. Add 100 mL of heptane. Close the test tube with stopper and shake well manually. The product does not dissolve in heptane.
3.0	Identification by: IR: Weigh accurately about 2.0 mg to 3.0 mg of Sitagliptin Phosphate CRS / working standard and 200.0 mg to 300.0 mg of dried potassium bromide, transfer into a mortar. Grind and mix it to fine powder with pestle. Spread it uniformly in suitable die and subject it to a pressure of about 6-8 bars. Release the pressure and remove the pellet with the spatula and mount on the pellet holder. The pellet should not be touched with hand. Record the IR spectrum over the range from 3800 and 650 cm ⁻¹ . Similarly this procedure followed to sample.

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	Overlay and compare the IR spectrum of sample with that of Sitagliptin Phosphate CRS / working standard. <i>Note: If the spectra obtained in the solid state shows differences, dissolve the sample to be examined and the reference sample separately in dehydrate alcohol, evaporate to dryness and record new spectra using the residues.</i> HPLC: Compare the retention time of major peak in the sample chromatogram to that in the standard chromatogram as obtained in Enantiomeric purity by HPLC. Phosphate test: Weigh accurately about 0.200 g of the sample to be examined, transfer in to 5 mL volumetric flask. Then dissolve and dilute to volume with water (40 mg / mL). Take above solution, add 5.0 mL of silver nitrate TS. A yellow precipitate is formed. Which is soluble in 2N nitric acid and in 6N ammonium hydroxide.	
4.0	Water content by KF: Transfer 40 mL of methanol in to the titration vessel and titrate with the Karl Fischer Reagent to the electrometric end point to consume any moisture that may be present. Weigh and transfer about 1.0 g of sample to be examined into the titration vessel of Karl Fischer apparatus. Taking precaution that complete sample is being transferred into the solvent and should not stick to the walls of the titration vessel. Mix it thoroughly by stirring and titrate with Karl Fischer Reagent to the end point stirring. <u>Calculation:</u> Calculate the % water content in sample by using the following formula. $\text{Water content (\%, w/w)} = \frac{\text{Titer value (mL)} \times \text{KF factor (mg/mL)} \times 100}{\text{Sample weight in (g)} \times 1000}$	

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S.No.	Test Procedure	
5.0	Residue on ignition: Ignite a suitable crucible at $600 \pm 50^{\circ}\text{C}$ for 30 minutes, allow to cool to room temperature for 30 minutes in a desiccator over silica gel. Weigh accurately and record the weight of the empty crucible (W_1). Weigh and transfer 1.0 g of the sample in a silica crucible and record the weight of the sample with crucible (W_2). Moisten the sample with 1.0 mL of sulfuric acid. Using a propane torch / Electric burner, Heat gently at first, at a temperature as low as practicable until the sample is thoroughly charred. Cool, then moisten the residue with 1.0 mL of sulfuric acid. Heat gently until white fumes are no longer evolved and ignite at $600 \pm 50^{\circ}\text{C}$ for 3 hours. Ensure that flames are not produced at any time during the procedure. Cool in a desiccator to room temperature, weigh the crucible with residue (W_3) and calculate the percentage of residue. Calculate the % of Residue on ignition of sample by using the following formula. $\text{Residue on ignition (\%, w/w)} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \times 100$ Unless otherwise specified, if the amount of the residue so obtained exceeds the limit specified in the specification, repeat the moistening with sulfuric acid, heating and igniting as before, using 30 min ignition period, until two consecutive weighing of the residue do not differ by more than 0.5 mg. Cool in a desiccator to room temperature, weigh the crucible with residue (W_4) and calculate the percentage of residue. Calculate the % of Residue on ignition of sample by using the following formula $\text{Residue on ignition (\%, w/w)} = \frac{(W_4 - W_1)}{(W_2 - W_1)} \times 100$	

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

6.0

Phosphate content by potentiometry:

Instrument Parameters:

Sensor : Aqueous
Start volume : 2 mL
Pause time : 60 sec
Titration rate : Optimal (medium)

Blank preparation and titration:

Transfer 50 mL of purified water into a potentiometric titration vessel, then dip the electrode into titration vessel and titrate against 0.1N sodium hydroxide solution, determine the end point potentiometrically.

Sample preparation and titration:

Weigh accurately and transfer about 0.3 g of sample to be examined into a potentiometric titration vessel. Add 50 mL of purified water, then stir about 1 minute until dissolve the sample. Then dip the Electrode into titration vessel and titrate against 0.1N sodium hydroxide solution and determine the end point potentiometrically.

Calculate the Phosphate Content (On Anhydrous basis) in sample solution by using below formula:

$$= \frac{(\text{Sample TV} - \text{Blank TV}) \times \text{Normality of 0.1N sodium hydroxide} \times 49 \times 100 \times 100}{\text{Weight of the sample in g} \times 1000 \times (100 - \text{Water content result of Sample in \%})}$$

Here,

TV = Titer value in mL

Note: The titer value for blank should be not more than 0.1 mL.

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S.No.

Test Procedure

7.0

Enantiomeric purity by HPLC:

Chromatographic Conditions:

Instrument : HPLC with UV detector
Column : Chiralpak, ADH, 250 mm x 4.6 mm, 5.0 μ m
Flow rate : 0.8 mL / minute
Wave length : 268 nm
Injection volume : 10 μ L
Column oven temperature : 35°C
Run time : 30.0 minutes
Elution mode : Isocratic

Preparation of mobile phase:

Transfer 600 mL of dehydrated ethanol, 400 mL of n-Heptane, 1.0 mL of diethyl amine and 1.0 mL of HPLC grade water into beaker and mix well. Then filter through 0.45 μ m filter paper.

Mobile phase stability: Mobile phase was stable up to 48 hours.

Preparation of diluent:

Take 900 mL of Methanol and 100 mL of HPLC grade water, mix well and filter through 0.45 μ m filter paper.

Preparation of S-Enantiomer standard stock solution:

Weigh accurately about 8.0 mg of S-Enantiomer standard, transfer into 100 mL volumetric flask. Add about 20 mL of diluent to dissolve and make up to mark with diluent and mix well.

Preparation of system suitability solution:

Weigh accurately about 80.0 mg of Sitagliptin phosphate CRS / working standard, transfer into 10 mL volumetric flask. Add about 3 mL of diluent to dissolve, add 2.0 mL of S-Enantiomer standard stock solution and make up to mark with diluent and mix.

Fig:01 shows the typical chromatogram of system suitability solution

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Test Procedure

Preparation of sample solution:

Weigh accurately about 80.0 mg of sample to be examined, transfer into 10 mL volumetric flask, add about 5 mL of diluent to dissolve and make up to mark with diluent and mix well.

Preparation of sensitivity solution:

Pipette out 1.0 mL of sample solution and transfer in to 100 mL volumetric flask. Make up to mark with diluent and mix well.

Pipette out 1.0 mL of the above solution and transfer in to 10 mL volumetric flask. Make up to mark with diluent and mix well.

Fig:02 shows the typical chromatogram of sensitivity solution

Solution stability: All solutions were stable for 24 hours at room temperature (25°C).

Injection sequence:

S. No.	Name of the injection	Number of injections
01	Blank solution	01 or 02
02	Sensitivity solution	01
03	System suitability solution	01
04	Blank solution	01
05	Sample solution	01

The retention time (RT), Relative retention time (RRT) values of S-enantiomer given in below table:

Name	About RT in minutes	RRT
S - enantiomer	12.66	0.90
Sitagliptin	15.38	1.00

Acceptance criteria for System suitability:

From system suitability solution: The resolution between S-Enantiomer and Sitagliptin peaks should be not less than 1.5.

From sensitivity solution: Signal to noise ratio of Sitagliptin should be not less than 10.

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Test Procedure

Calculation: Calculate % of S-enantiomer in the portion of Sitagliptin phosphate taken:

$$= (r_u / r_t) \times 100$$

Here,

r_u = Peak response of S-enantiomer from the sample chromatogram.

r_t = Sum of peak responses of S-enantiomer and Sitagliptin from the Sample chromatogram.

The Limit Of Detection (LOD), Limit Of Quantitation (LOQ) values of S-enantiomer given in below table:

Name	LOQ conc. in (%) w.r.t sample	LOD conc. in (%) w.r.t sample
S - enantiomer	0.026	0.0086

8.0

Organic impurities by HPLC:

Chromatographic Conditions:

Instrument : HPLC with UV detector
 Column : Supleco, Discovery @cyano (150 x 4.6) mm, 5.0 μ m
 Cat#:59356-U.
 Flow rate : 1.0 mL / minute
 Wave length : 205 nm
 Injection volume : 20 μ L
 Run Time : 35.0 minutes
 Column oven Temperature: 30°C
 Elution mode : Isocratic

Preparation of Buffer solution:

Weigh accurately about 1.36 g of Potassium di hydrogen ortho phosphate, transfer into beaker containing 1000 mL of HPLC grade water, sonicate and dissolve. Then adjust pH to 2.0 ± 0.05 with dilute Ortho phosphoric acid. Degas to 5 minutes at ambient temperature and filter through 0.22 μ m filter paper.

Preparation of mobile phase:

Degas the mixture of 150 mL of Acetonitrile, 850 mL of buffer solution and filter through 0.22 μ m filter paper.

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	<u>Mobile phase stability:</u> Mobile phase was stable up to 62 hours. <u>Preparation of dilute ortho phosphoric acid solution (0.1 % v/v):</u> Transfer 1.0 mL of ortho phosphoric acid into 1000 mL volumetric flask and make up to mark with HPLC grade water. <u>Preparation of diluent:</u> Degas the mixture of 50 mL of acetonitrile, 950 mL of dilute ortho phosphoric acid solution (0.1 % v/v) and filter through 0.22 µm filter paper. <u>Preparation of system suitability solution:</u> Weigh accurately about 1.0 mg of sodium stearyl fumarate and 10.0 mg of Sitagliptin phosphate USP CRS / working standard, transfer in to a vial, tightly close the vial and keep at 80°C for about 30 hours to generate a fumarate adduct of sitagliptin. Transfer the contents of vial to 100 mL volumetric flask using a small amount of diluent and makeup with diluent and stir for 1 hour. Centrifuge a portion of the solution for 10 minutes or until the solution is clear, and use the supernatant. <u>Note:</u> The chemical name of fumarate adduct of sitagliptin is 2-[[[(R)-4-oxo-4-[3-(trifluoromethyl)-5,6-dihydro-[1,2,4]triazolo[4,3-a]pyrazine-7(8H)-yl]-1-(2,4,5-trifluoro phenyl) butan-2-yl]amino]succinic acid] <u>Fig: 03 shows the typical chromatogram of system suitability solution.</u> <u>Preparation of standard solution:</u> Weigh accurately about 25.0 mg of Sitagliptin phosphate CRS / working standard, transfer in to 25 mL volumetric flask, Add about 5 mL of diluent to dissolve and make up to mark with diluent and mix well. Pipette out 1.0 mL of above solution and transfer into 100 mL volumetric flask. Make up to mark with diluent and mix well. Pipette out 1.0 mL of the above solution and transfer into 100 mL volumetric flask. Make up to mark with diluent and mix well. <u>Fig: 04 shows the typical chromatogram of standard solution.</u>

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S.No.

Test Procedure

Preparation of sample solution:

Weigh accurately about 50.0 mg of sample to be examined, transfer into 50 mL volumetric flask, add about 20 mL of diluent to dissolve and make up to mark with diluent and mix well.

Pipette out 5.0 mL of above solution and transfer into 50 mL volumetric flask. Make up mark with diluent and mix well.

Solution stability: All the solutions were stable up to 24 hours at room temperature.

Injection sequence:

S. No.	Name of the injection	No. of injections
01	Blank solution	01 or 02
02	System suitability solution	01
03	Blank solution	01
04	Standard solution	01
05	Sample solution	01

The retention time (RT) values of Sitagliptin and fumarate adduct given in below table:

Name	About RT in minutes
Sitagliptin	5.28
Fumarate adduct	6.24

Acceptance criteria for System suitability:

From system suitability solution: The resolution between Sitagliptin and fumarate adduct is should be not less than 1.5.

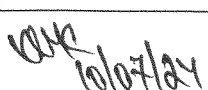
Note:

- The relative retention times for Sitagliptin and fumarate adduct of sitagliptin are 1.0 and 1.2.
- Disregard any peak below 0.05 %.

Calculation: Calculate the % of each impurity in the portion of Sitagliptin phosphate taken.

% any individual impurity

$$= \frac{AT}{AS} \times \frac{Std. wt. (mg)}{25} \times \frac{1}{100} \times \frac{1}{100} \times \frac{50}{Sample wt. (mg)} \times \frac{50}{5} \times 100$$

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% of total impurities

$$= \frac{\text{AZ}}{\text{AS}} \times \frac{\text{Std. wt. (mg)}}{25} \times \frac{1}{100} \times \frac{1}{100} \times \frac{50}{\text{Sample wt (mg)}} \times \frac{50}{5} \times 100$$

Here,

AT = Area of any individual impurity obtained from sample chromatogram.

AS = Area of Sitagliptin peak from standard solution chromatogram.

AZ = Sum of peak areas of all impurities other than main peak obtained from sample chromatogram.

Limit Of Detection (LOD), Limit Of Quantitation (LOQ) values of Sitagliptin given in below table:

Name	LOQ conc. in (%) w.r.t sample	LOD conc. in (%) w.r.t sample
Sitagliptin	0.03012	0.0099

9.0 Assay by HPLC:

Chromatographic Conditions:

Instrument : HPLC with UV detector

Column : Make: Supleco, Discovery @cyano (150 x 4.6) mm, 5.0 µm. Cat #: 59356 U

Flow rate : 1.0 mL / minute

Wave length : 205 nm

Injection volume : 20 µL

Run Time : 15.0 minutes

Column oven Temperature: 30°C

Elution mode : Isocratic

Preparation of Buffer solution:

Weigh accurately about 1.36 g of Potassium di hydrogen ortho phosphate, transfer into beaker containing 1000 mL of HPLC grade water. Sonicate and dissolve. Then adjust pH to 2.0 ± 0.05 with dilute ortho phosphoric acid. Degas to 5 minutes at ambient temperature and filter with 0.22 µm filter paper.

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S.No.	Test Procedure
	<u>Preparation of dilute ortho phosphoric acid solution (0.1% v/v):</u> Pipette out 1.0 mL of ortho phosphoric acid and transfer into 1000 mL volumetric flask, make up to mark with HPLC grade water and mix well. <u>Preparation of Mobile phase:</u> Degas the mixture of 150 mL of acetonitrile, 850 mL of buffer solution and filter through 0.22 µm filter paper. <u>Mobile phase stability:</u> Mobile phase was stable up to 48 hours. <u>Preparation of diluent:</u> Degas the mixture of 50 mL acetonitrile and 950 mL of dilute ortho phosphoric acid solution (0.1% v/v). Filter through 0.22 µm filter paper. <u>Preparation of standard solution:</u> Weigh accurately about 50.0 mg of Sitagliptin phosphate CRS / working standard, transfer into 50 mL volumetric flask, add about 20 mL of diluent, mix well and make up to mark with diluent and mix well. Pipette out 5.0 mL of above solution and transfer into 50 mL volumetric flask. Make up to mark with diluent and mix well. <u>Fig: 05 shows the typical chromatogram of standard solution.</u> <u>Preparation of sample solution - 01:</u> Weigh accurately about 50.0 mg of sample to be examined and transfer into 50 mL volumetric flask, add about 20 mL of diluent, mix well and make up to mark with diluent and mix well. Pipette out 5.0 mL of above solution and transfer into 50 mL volumetric flask. Make up to mark with diluent and mix well. <u>Preparation of sample solution - 02:</u> Weigh accurately about 50.0 mg of sample to be examined and transfer into 50 mL volumetric flask, add about 20 mL of diluent, mix well and make up to mark with diluent and mix well. Pipette out 5.0 mL of above solution and transfer into 50 mL volumetric flask. Make up to mark with diluent and mix well.

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Test Procedure

Solution stability: All the solutions were stable up to 24 hours at room temperature.

Injection sequence:

S. No.	Name of the injection	No. of injections
01	Blank solution	01 or 02
02	Standard solution	05
03	Sample solution preparation - 01	01
04	Sample solution preparation - 02	01

Note: The Retention time of Sitagliptin peak is about 5.37 minutes

Acceptance criteria for System suitability:

From standard solution: % RSD of peak areas of Sitagliptin from five replicate injections should be not more than 0.73.

Calculation: Calculate % of assay (On anhydrous and solvent free basis) in sample solution by using below formula:

$$\% \text{ Assay} = \frac{AT}{AS} \times \frac{C_s}{C_T} \times \frac{100}{[100 - (WC + \text{Total Organic solvents in } \%)]} \times P$$

Here,

AT = Area of principal peak obtained from sample chromatogram.

AS = Average area of principal peak obtained from standard chromatograms.

C_s = Concentration of Sitagliptin phosphate RS in the standard solution (mg / mL)

C_T = Concentration of Sitagliptin phosphate in the sample solution (mg / mL)

P = % Assay of Sitagliptin phosphate standard on as is basis or Potency.

WC = % of Water content result of sample.

$$\% \text{ of Total Organic solvents} = \frac{\text{Total ppm of Residual solvents}}{10000}$$

Note: Use above calculation for sample preparation - 01 and 02. Individually should meet the specification limit. Then report the average result.

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Test Procedure

10.0

Residual solvents by GC:

Chromatographic conditions:

Instrument : The gas chromatograph equipped with flame ionization detector with headspace sampler.

Column information

Column : DB-624

Dimensions : 60 m x 0.53 mm ID, 3.0 µm.

Temperature program:

Rate (°C/min)	Temperature (°C)	Hold time (min)
--	40	15.0
20	220	15.0

Total run time : 39.0 minutes

Detector parameters:

Detector : Flame ionization detector

Detector temperature : 250°C

Makeup flow : 35.0 mL / minute

Hydrogen flow : 40.0 mL / minute

Air flow : 400.0 mL / minute

Injection parameters:

Injection mode : Split

Split Ratio : 1:4

Column flow : 2.50 mL / minute

Carrier gas : Helium

Note:

Injection temperature is not applicable for Shimadzu GC -2010 plus and Nexis GC - 2030 with HS-20 model. Because the column inlet is directly connected to transfer line. Hence based on vendor (Shimadzu-spincoteam) recommendation injection temperature for other make instruments will be considered as 220°C.

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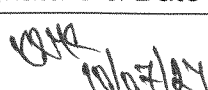
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S.No.	Test Procedure									
	<u>Auto sampler Shimadzu (HS-20) parameters:</u> Oven temperature : 90°C Sample line temperature : 110°C Transfer line temperature : 110°C Equilibrating time : 30.0 minutes Pressurization time : 3.0 minutes Pressure Equilibration time : 0.10 minute Load time (Loop fill) : 0.50 minutes Load Equilibration time : 0.10 minute Injection time : 0.5 minutes Shaking level : 1 Multi injection count : 1 Needle flush time : 5.0 minutes GC Cycle time : 60.0 minutes Pressurize gas pressure : Should be 80.0 kpa <u>Chromatographic conditions for Agilent (7890B GC):</u> Instrument : The gas chromatograph equipped with flame ionization detector with headspace sampler. <u>Column information</u> Column : DB-624 Dimensions : 60 m x 0.53 mm ID, 3.0 µm. <u>Temperature program:</u> <table><tr><th>Rate (°C/min)</th><th>Temperature (°C)</th><th>Hold time (min)</th></tr><tr><td>--</td><td>40</td><td>15.0</td></tr><tr><td>20</td><td>220</td><td>15.0</td></tr></table> Total run time : 39.0 minutes <u>Detector parameters:</u> Detector : Flame ionization detector.	Rate (°C/min)	Temperature (°C)	Hold time (min)	--	40	15.0	20	220	15.0
Rate (°C/min)	Temperature (°C)	Hold time (min)								
--	40	15.0								
20	220	15.0								

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S.No.	Test Procedure
	Detector temperature : 250°C Makeup flow : 35.0 mL/ minute Hydrogen flow : 40.0 mL/ minute Air flow : 400.0 mL/ minute <u>Injection parameters:</u> Injection temperature : 180°C Injection mode : Split Split Ratio : 1:4 Column flow : 2.50 mL/ minute Carrier gas : Helium <u>Auto sampler parameters (7697A Head space sampler):</u> Oven temperature : 90°C Loop temperature : 110°C Transfer line temperature : 110°C Vial Equilibrating time : 30.0 minutes GC Cycle time : 60.0 minutes Injection time : 0.50 minutes Fill pressure : 12 psi Loop fill pressure : 7 psi Loop equilibrating time : 0.05 minutes Diluent : N,N-Dimethyl acetamide <u>Preparation of blank solution:</u> Take 2.0 mL of Diluent and 1.0 mL of HPLC grade water, transfer into a headspace vial. Place the septa and immediately crimp the vial. <u>Preparation of Chloroform standard stock solution:</u> Transfer 60.0 mg of Chloroform Standard into 50 mL volumetric flask containing about 20 mL of diluent, mix well and dilute to volume with diluent.

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Test Procedure

Preparation of standard stock solution:

Transfer 300.0 mg of Methanol Standard, 500.0 mg of Acetone Standard, 500.0 mg of Isopropyl alcohol Standard, 41.0 mg of Acetonitrile Standard, 60.0 mg of Dichloromethane Standard, 500.0 mg of Methyl tertiary butyl ether Standard, 500.0 mg of n-Butanol Standard, 89.0 mg of Toluene Standard and 5.0 mL of Chloroform standard stock solution into 50 mL volumetric flask containing about 20 mL of diluent, mix well and make up to mark with diluent and mix well.

Preparation of standard solution:

Transfer 5.0 mL of standard stock solution into 100 mL volumetric flask containing about 40 mL of diluent, mix well and make up to mark with diluent and mix well. Transfer 2.0 mL of standard solution and 1.0 mL HPLC grade water into 20 mL Headspace vial and immediately crimp the vial.

Note: In similarly prepare required number of standard solution vials.

Fig:06 shows the typical chromatogram for standard solution.

Preparation of Sample solution:

Weigh accurately about 200.0 mg of sample to be examined, transfer into 20 mL headspace vial. Add 2.0 mL of diluent and 1.0 mL of HPLC grade water, then immediately crimp the vial.

Solution stability: All the standard and Sample solutions were stable up to 24 hours at room temperature.

Injection sequence:

S. No.	Name of the injection	No. of injections
01	Blank solution	01 or 02
02	Standard solution	06
03	Blank solution	01
04	Sample solution	01

Acceptance criteria for System suitability:

Form standard solution injection- 01: The resolution between any two closest specified solvent peaks should be not less than 2.0.

Form standard solution: % RSD for the peak areas of all specified solvents obtained from six injections should be not more than 15.0.

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 Lee Pharma Limited Unit-I, Gaddapotharam	SPECIFICATIONS AND TEST PROCEDURE		
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Product Name : Sitagliptin Phosphate			
S.No.	Test Procedure		
	<u>Calculation:</u> <u>Calculate the ppm content of all known solvents (except Chloroform) from sample solution by using below formula:</u> $\text{ppm content} = \frac{A_T}{A_S} \times \frac{W_S}{50} \times \frac{5}{100} \times \frac{2}{3} \times \frac{3}{W_U} \times \frac{P}{100} \times 10^6$ Here, A_T = Peak area of respective specified solvent from Sample chromatogram. A_S = Average peak area of respective specified solvent from six injections of Standard chromatograms. W_S = Weight of respective solvent standard in mg W_U = Weight of sample taken in mg P = % Purity of respective solvent standard <u>Calculate the ppm content of Chloroform from sample solution by using below formula:</u> $\text{ppm content} = \frac{A_T}{A_S} \times \frac{W_S}{50} \times \frac{5}{50} \times \frac{5}{100} \times \frac{2}{3} \times \frac{3}{W_U} \times \frac{P}{100} \times 10^6$ Here, A_T = Peak area of Chloroform from Sample chromatogram. A_S = Average peak area of Chloroform from six injections of Standard chromatograms. W_S = Weight of Chloroform standard in mg W_U = Weight of sample in mg P = % Purity of Chloroform standard		

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

Retention Time (RT), Limit Of Quantitation (LOQ), Limit Of Detection (LOD) values of each specified solvent given in below table:

Name of the solvent	About RT in minutes	LOQ Conc. in (ppm) w.r.t sample	LOD Conc. in (ppm) w.r.t sample
Methanol	9.48	51.55	17.01
Acetone	15.07	10.01	3.30
Isopropyl alcohol	15.78	25.07	8.27
Acetonitrile	16.26	26.83	8.85
Dichloromethane	16.93	14.64	4.83
Methyl tertiary butyl ether	17.90	1.50	0.50
Chloroform	21.61	19.24	6.35
n-Butanol	24.19	45.20	14.92
Toluene	27.20	3.57	1.18

11.0 Acetic acid content by HPLC:

Chromatographic Conditions:

Instrument	:	HPLC with UV detector
Column	:	Inertsil ODS-3 (250 x 4.6) mm, 5.0 µm
Wavelength	:	210 nm
Flow rate	:	1.0 mL /minute
Injection volume	:	10 µL
Runtime	:	45 minutes
Column oven temperature	:	30°C
Auto sampler temperature	:	5°C
Mode	:	Gradient
Diluent	:	Methanol
Needle wash solution	:	HPLC grade Water: Acetonitrile (1:1) (% , v/v)

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

Gradient Program:

Time (minutes)	% Mobile phase - A	% Mobile phase - B
0.01	100	0
10.00	100	0
13.00	10	90
29.00	10	90
30.00	100	0
45.00	100	0
45.01	Stop	

Preparation of Mobile phase - A:

Transfer 1.0 mL of ortho phosphoric acid into beaker containing 1000 mL of HPLC grade water, mix well. Filter through 0.45 µm filter paper.

Preparation of Mobile phase - B:

Should be use acetonitrile.

Mobile phase stability: Mobile phase was stable up to 48 hours.

Preparation of Acetic acid Standard Stock solution:

Weigh accurately about 50.0 mg of Acetic acid standard into 100 mL volumetric flask, add about 40 mL of diluent to dissolve and make up to mark with diluent and mix.

Preparation of standard solution:

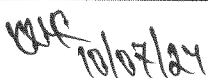
Pipette out 5.0 mL of Acetic acid standard stock solution and transfer into 50 mL volumetric flask containing about 20 mL of diluent. Mix well and make up to mark with diluent and mix.

Fig: 07 shows the typical chromatogram of standard solution.

Preparation of sample solution:

Weigh accurately about 100.0 mg of sample to be examined and transfer into 10 mL volumetric flask, add about 5 mL of diluent to dissolve and make up to mark with diluent and mix.

Solution stability: All Standard and sample solutions were stable up to 24 hours at room temperature and at 5°C.

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

Injection Sequence:

S. No.	Name of the injection	No. of injections
01	Blank solution	01 or 02
02	Standard solution	6
03	Blank solution	1
04	Sample solution	1

Note: The Retention time of acetic acid peak is about 6.42 minutes.

Integrate only acetic acid peak in sample chromatogram.

Acceptance criteria for system suitability: From standard solution: % RSD for the peak areas of acetic acid from six replicate injections should be not more than 5.0.

Calculate the ppm content of Acetic acid in sample solution by using below formula

$$= \frac{A_u}{A_s} \times \frac{W_s \times f}{100} \times \frac{5}{50} \times \frac{10}{W_u} \times 10^6$$

Here,

A_u = Area of Acetic acid peak in the sample chromatogram

A_s = Average area of Acetic acid in standard chromatograms

W_s = Weight of Acetic acid standard in mg

W_u = Weight of sample in mg

f = Purity factor for Acetic acid standard ($f = \text{Purity} / 100$)

10^6 = Conversion factor to ppm

Limit Of Quantitation (LOQ), Limit Of Detection (LOD) values of Acetic acid solvent given in below table:

Name	LOQ Conc. in (ppm) w.r.t sample	LOD Conc. in (ppm) w.r.t sample
Acetic acid	1317	434

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

12.0 NTPy / NTTP Nitrosamine impurity content by LC - MS:

Chromatographic conditions:

Instrument : LCMS
Column : Shimadzu, Shim pack GIST C18
(250 x 4.6) mm, 5.0 µm
Flow rate : 0.7 mL/minute
Injection volume : 20 µL
Column oven temperature: 45°C
Auto sampler temperature: 15°C
Run Time : 28.0 minutes

Gradient program:

Time (minutes)	Mobile phase-A (%)	Mobile phase-B (%)
0.01	80	20
2.00	80	20
7.00	60	40
12.00	45	55
15.00	05	95
19.00	05	95
20.00	80	20
28.00	80	20
28.01	Stop	

Preparation of Mobile phase - A:

Pipette out 1.0 mL of formic acid and transfer into beaker containing 1000 mL of LCMS grade water, mix well and sonicate to degas. Then filter through 0.45 µm filter paper.

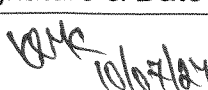
Preparation of Mobile Phase - B:

Pipette out 1.0 mL of formic acid and transfer into beaker containing 1000 mL of LCMS grade methanol, mix well and sonicate to degas. Then filter through 0.45 µm filter paper.

Preparation of Diluent: Ethyl acetate use as diluent.

Preparation of needle wash solution: Methanol should be use as need wash solution.

Preparation of blank solution: Diluent use as blank solution.

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Product Name : Sitagliptin Phosphate

S.No. Test Procedure

Interface parameters:

Interface : ESI
Acquisition Mode : MRM
Polarity : Positive
Nebulizing Gas Flow : 2.00 L/minute
Interface Temperature : 200°C
DL Temperature : 150°C
Heat Block temperature : 200°C
Drying Gas Flow : 10.0 L/minute
Heat gas flow : 10.0 L / minute

MRM Transitions:

Name	Precursor m/z	Product m/z	Q1 Pre Bias (V)	Collision Energy	Q3 Pre Bias (V)
NTPy / NTPP	222.10	192.15	-16	-10	-20

MS program Valve Position:

Initial valve position:1

Time	Command	Valve
0.10	FCV2=	1
10.00	FCV2=	0
25.00	FCV2=	1

Preparation of NTPy/NTPP standard stock solution:

Weigh accurately and transfer 9.2 mg of NTPy/NTPP standard into 15 mL tarson tube, then add 5.0 mL of methanol. Mix well to dissolve and make up with methanol up to 10 mL mark.

Preparation of standard stock solution - 1:

Pipette out 0.8 mL of NTPy/NTPP standard stock solution and transfer into 50 mL tarson tube containing about 20 mL of methanol, mix well and make up to mark with methanol.

Preparation of standard stock solution-2:

Pipette out 0.5 mL of standard stock solution-1 and transfer into 50 mL tarson tube containing about 20 mL of diluent, mix well and make up to mark with diluent.

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Product Name : Sitagliptin Phosphate

S.No.

Test Procedure

Preparation of standard solution:

Pipette out 1.0 mL of standard stock solution-2 and transfer into 50 mL tarson tube containing about 20 mL of diluent, mix well and make up to mark with diluent.

Fig: 08 shows the typical chromatogram of standard solution.

Solution stability: Standard solution was stable up to 29 Hours at room temperature (25°C) and at 38°C.

Preparation of sensitivity solution (0.03 ppm) w.r.t sample:

Pipette out 0.10 mL of standard stock solution-2 and transfer into 50 mL tarson tube containing about 20 mL of diluent, mix well and make up to mark with diluent.

Fig: 09 shows the typical chromatogram of sensitivity solution.

Preparation of sample solution:

Weigh accurately about 100.0 mg of sample to be examined and transfer into 15 mL centrifuged tube, add 10 mL of diluent and vortex for 60 seconds. Then sonicate for 5 minutes and mix well on vortex mixer for about 60 seconds. Then Centrifuge the sample solution for 10 minutes at 4500 rpm speed. Filter the sample solution through 0.22 µm Nylon syringe filters.

Solution stability: Sample solution was stable up to 25 Hours at room temperature (25°C) and at 35°C.

Injection sequence:

S.No.	Name of the injection	No. of injections
01	Blank solution	01 or 02
02	Sensitivity solution	01
03	Standard solution	01
04	Blank solution	01
05	Standard solution	06
06	Blank solution	01
07	Sample solution	01

Note: The Retention time of NTPy / NTTP peak is about 8.72 minutes

Acceptance criteria for system suitability:

From Sensitivity solution: The S / N ratio for the NTPy / NTTP peak is should be not less than 10.0

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	From standard solution: % RSD for the peak areas of NTPy / NTTP obtained from six replicate injections is should be not more than 20.0. Calculate ppm content of NTPy/NTTP from sample solution by using below formula: $\text{ppm content} = \frac{A_u}{A_s} \times \frac{W_s}{10} \times \frac{1}{50} \times \frac{0.5}{50} \times \frac{1.0}{50} \times \frac{10}{W_u} \times \frac{P}{100} \times 10^6$ Here, A_u = Peak area of NTPy/NTTP impurity from sample chromatogram. A_s = Average peak area of NTPy/NTTP impurity from six standard chromatograms. W_s = Weight of NTPy/NTTP impurity standard taken in mg. W_u = Weight of the sample taken in mg. P = Potency/purity of NTPy/NTTP impurity standard. Corrected peak area = (Peak area of area respective impurity from standard/ sample)- (Peak area of respective impurity from blank) Retention Time (RT), Limit of Detection (LOD) and Limit of Quantitation (LOQ) values of NTPy given in below table: <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Name</th> <th>LOD conc. in ppm w.r.t sample</th> <th>LOQ conc. in ppm w.r.t sample</th> </tr> </thead> <tbody> <tr> <td>NTPy / NTTP</td> <td>0.0120</td> <td>0.0363</td> </tr> </tbody> </table>		Name	LOD conc. in ppm w.r.t sample	LOQ conc. in ppm w.r.t sample	NTPy / NTTP	0.0120	0.0363
Name	LOD conc. in ppm w.r.t sample	LOQ conc. in ppm w.r.t sample						
NTPy / NTTP	0.0120	0.0363						

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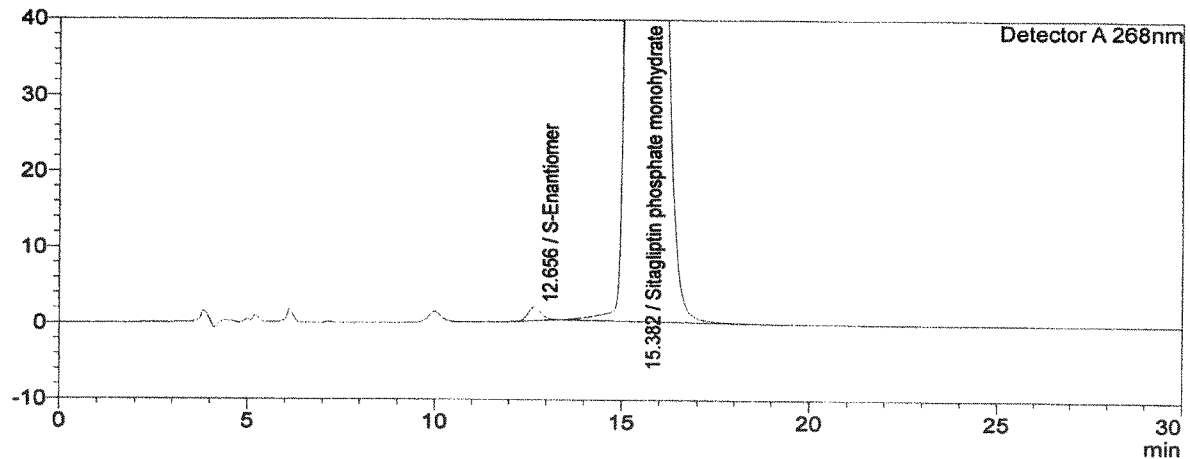
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Product Name : Sitagliptin Phosphate

Fig:01 (Enantiomeric purity by HPLC)

Typical chromatogram of system suitability solution



Peak#	Name	Ret. Time	Area	Area%	RRT
1	S-Enantiomer	12.656	44927	0.1993	0.82
2	itagliptin phosphate monohydra	15.382	22497595	99.8007	1.00
Total			22542522	100.0000	

Resolution(USP)
3.231

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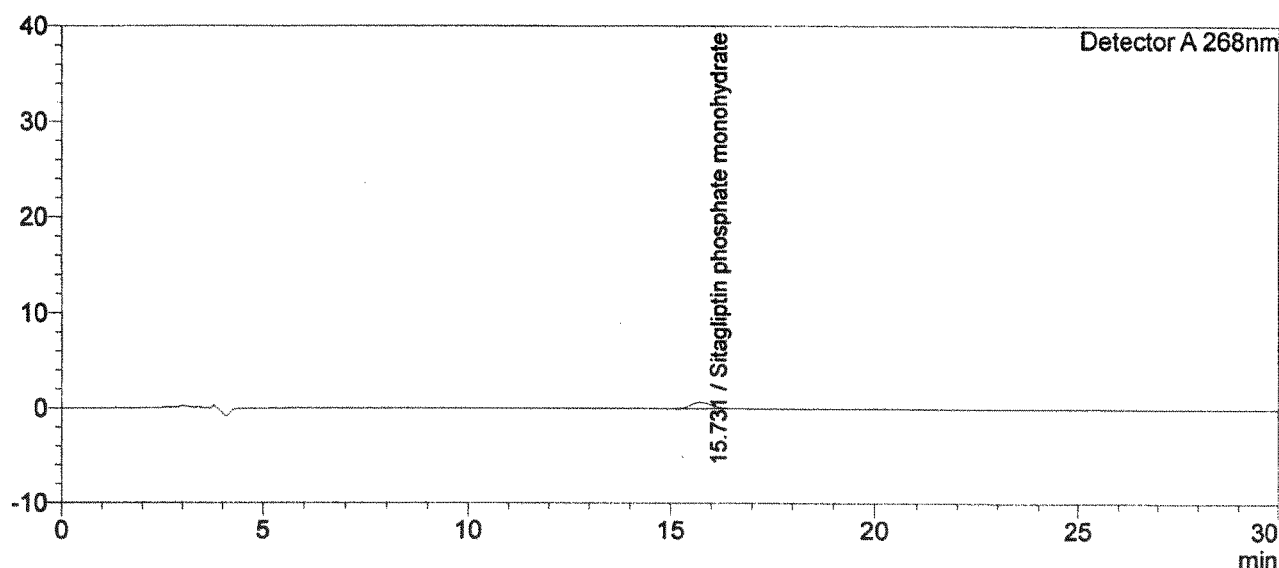
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Product Name : Sitagliptin Phosphate

Fig:02 (Enantiomeric purity by HPLC)

Typical chromatogram of sensitivity solution



Peak#	Name	Ret. Time	Area	Area%	Height	RRT
1	itagliptin phosphate monohydrate	15.731	22613	100.0000	672	--
Total			22613	100.0000	672	

S/N
82.13

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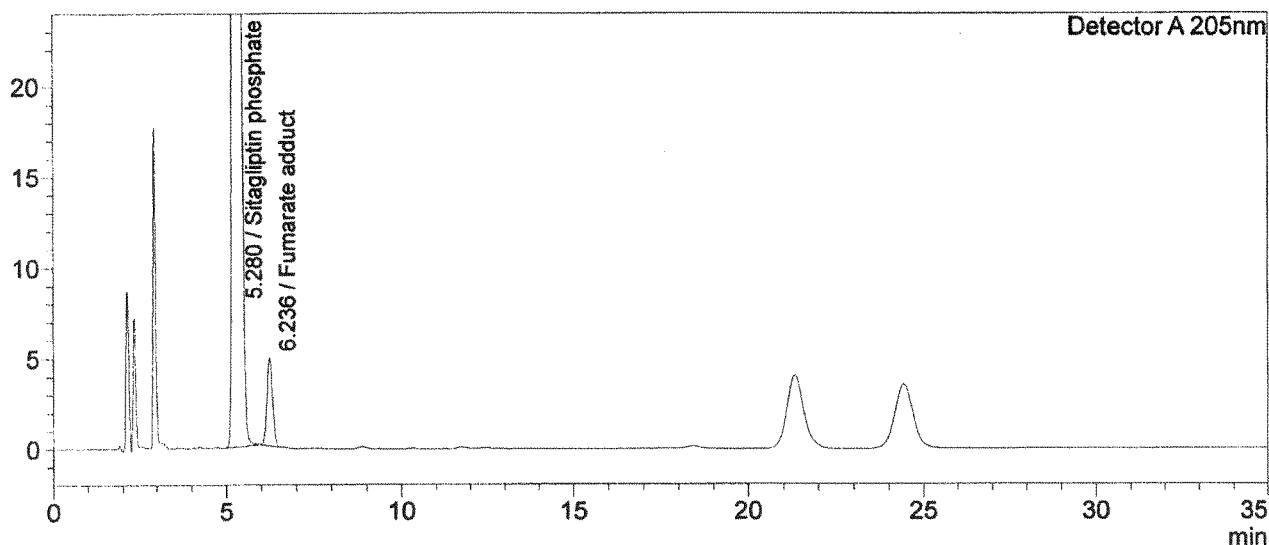
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Product Name : Sitagliptin Phosphate

Fig:03 (Organic impurities by HPLC)

Typical chromatogram of system suitability solution



Detector A 205nm

Peak#	Name	Ret. Time	Area	Area%	Height
1	Sitagliptin phosphate	5.280	3316109	98.385	349909
2	Fumarate adduct	6.236	54437	1.615	4870
Total			3370546	100.000	354779

Resolution(USP)	Relative Retention Time
--	0.847
3.403	1.000

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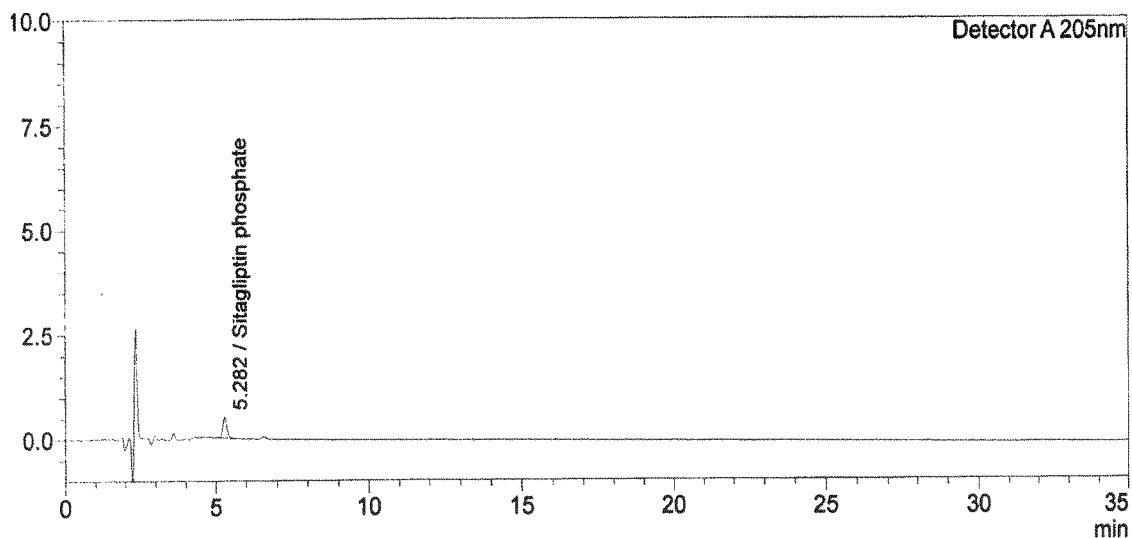
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Product Name : Sitagliptin Phosphate

Fig:04 (Organic impurities by HPLC)
Typical chromatogram of standard solution



Detector A 205nm

Peak#	Name	Ret. Time	Area	Area%	Height
1	Sitagliptin phosphate	5.282	4620	100.000	513
Total			4620	100.000	513

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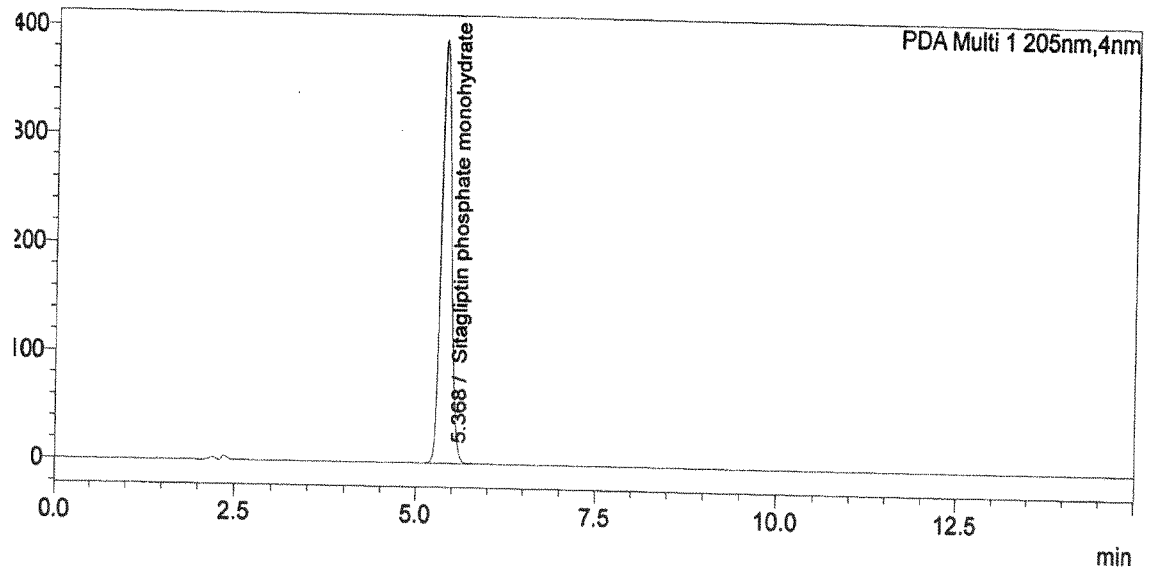
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Fig:05 (Assay by HPLC)

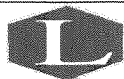
Typical chromatogram of standard solution



Peak#	Name	Ret. Time	Area	Area%
1	Sitagliptin phosphate monohydrate	5.368	3761956	100.00
Total			3761956	100.00

Number of Theoretical Plate(USP)	Tailing Factor
6413	1.351

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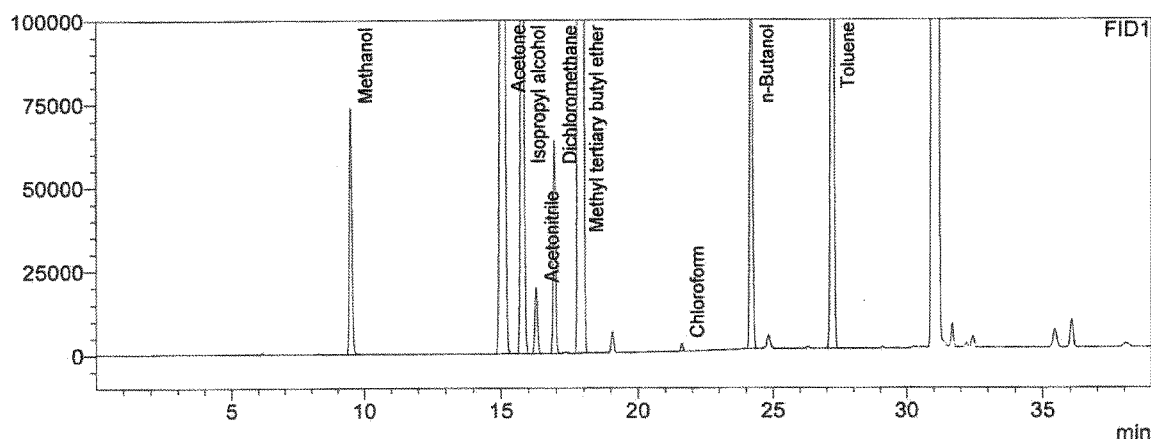
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Product Name : Sitagliptin Phosphate

Fig:06 (Residual solvents by GC)

Typical chromatogram of standard solution



Peak#	Ret. Time	Area	Height	Area%	Name	Resolution(USP)
1	9.483	485155	73411	1.364	Methanol	--
2	15.072	4128247	437591	11.604	Acetone	26.419
3	15.780	1627130	218274	4.574	Isopropyl alcohol	3.205
4	16.264	128547	19503	0.361	Acetonitrile	2.646
5	16.929	379945	63438	1.068	Dichloromethane	3.998
6	17.897	25251646	3465557	70.977	Methyl tertiary butyl ether	5.459
7	21.606	14129	2436	0.040	Chloroform	21.219
8	24.189	868117	131212	2.440	n-Butanol	15.654
9	27.203	2694100	397418	7.573	Toluene	16.972
Total		35577016	4808841	100.000		

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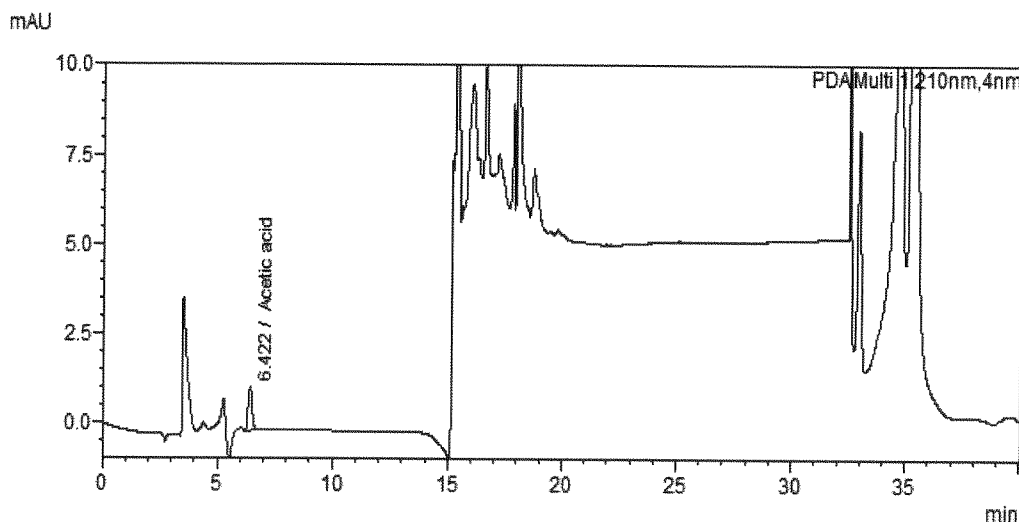
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Product Name : Sitagliptin Phosphate

Fig:07 (Acetic acid content by HPLC)
Typical chromatogram of standard solution



<Peak Table>

PDA Ch1 210nm

Peak#	Ret. Time	Area%	Area	Name
1	6.422	100.00	15085	Acetic acid
Total		100.00	15085	

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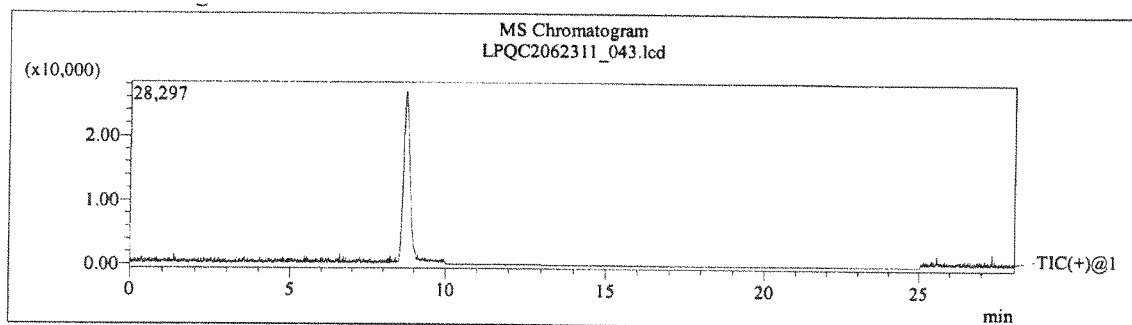
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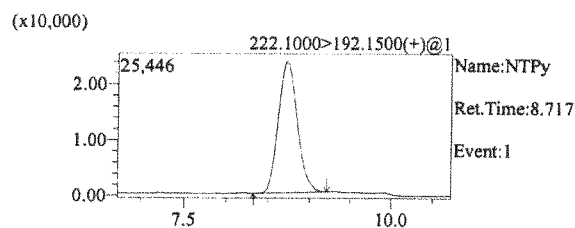
Fig:08 (NTPy / NTTP Nitrosamine impurity content by LC - MS)

Typical chromatogram of standard solution



MS Quantitative Graph

MS Quantitative Graph
*LPQC2062311_043.lcd



MS Quantitative Table

MS Quantitative Table
LPQC2062311_043.lcd

ID#	Name	Ret. Time	m/z	Area
1	NTPy	8.717	222.10>192.15	375732
Total				375732

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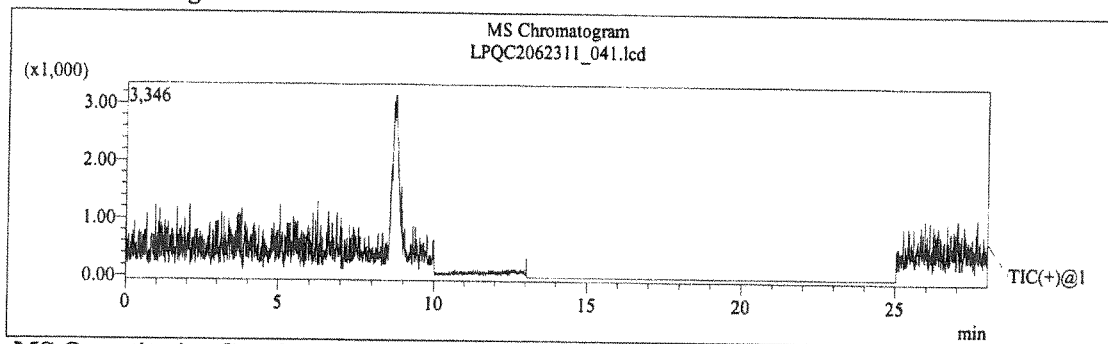
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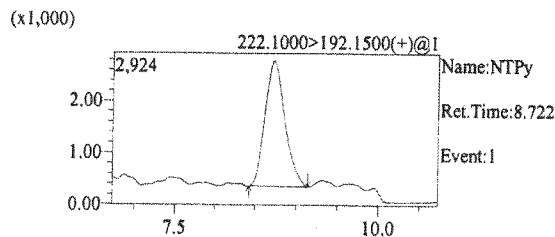
Fig:09 (NTPy / NTTP Nitrosamine impurity content by LC - MS)

Typical chromatogram of sensitivity solution



MS Quantitative Graph

MS Quantitative Graph
*LPQC2062311_041.lcd



MS Quantitative Table

MS Quantitative Table
LPQC2062311_041.lcd

ID#	Name	Ret. Time	m/z	Area	S/N
1	NTPy	8.722	222.10>192.15	41170	18.82
Total				41170	

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C. REVISION HISTORY:

Rev. No.	Effective Date	Details of Revision	Ref. Change Control No.
00	31.08.2020	➤ First issue	CCM20060
01	23.09.2020	➤ Residual solvents by GC and Acetic acid content by HPLC analytical methods were updated in line with analytical method validation.	CCQ20167
02	09 NOV 2020	➤ In storage condition mode, storage at room temperature condition is established in numerical values as 15°C to 30°C. ➤ In Residual solvents by GC, solution Stability temperature conditions included as per Analytical method validation and diluent details updated. ➤ In acetic acid content by HPLC, solution stability temperature and LOD & LOQ values included as per analytical method validation.	CCQ20193
03	01 APR 2022	➤ Storage condition revised to "Store in well – closed containers at 25°C, excursion is permitted between 15°C to 30°C" from "Store in a well closed container at 15°C to 30 °C " ➤ Sitagliptin Phosphate chemical name (IUPAC) typo error corrected. ➤ Method reference numbers included for each test parameter in specification.	CCQ22069

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		K. Siva Santhosh Kumar	Asst. Manager - QA

 Lee Pharma Limited Unit-I, Gaddapotharam		SPECIFICATIONS AND TEST PROCEDURE	
		STP No. : STP/FP/SP/002-08	Page No. : 38 of 41
Product Name : Sitagliptin Phosphate		Department : QUALITY CONTROL	
Rev. No.	Effective Date	Details of Revision	Ref. Change Control No.
03	01 APR 2022	➤ Identification by phosphate test updated as per current USP monograph. ➤ In enantiomeric purity by HPLC, S-enantiomer acceptance criteria revised to 'NMT 0.50' from 'NMT 0.5'. ➤ In Organic impurities by HPLC, Total impurities acceptance criteria revised to 'NMT 0.50' from 'NMT 0.5'. ➤ Water content by KF acceptance criteria updated to 'Between 3.3 and 3.7' from 'Between 3.3 to 3.7'. ➤ Residue on ignition and Phosphate content by potentiometry (On anhydrous basis) test parameters included. ➤ Mobile phase stability, solution stability, typical chromatograms of sensitivity solution and system suitability solution, injection sequence harmonization, Limit Of Detection (LOD), Limit Of Quantitation (LOQ) values of S-enantiomer incorporated in Enantiomeric purity by HPLC method of analysis as per analytical method validation.	CCQ22069

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 Lee Pharma Limited Unit-I, Gaddapotharam		SPECIFICATIONS AND TEST PROCEDURE	
		STP No. : STP/FP/SP/002-08	Page No. : 39 of 41
Product Name : Sitagliptin Phosphate		Department : QUALITY CONTROL	
Rev. No.	Effective Date	Details of Revision	Ref. Change Control No.
03	01 APR 2022	➤ Mobile phase stability, solution stability, typical chromatograms of standard solution and system suitability solution, injection sequence harmonization, Limit Of Detection (LOD), Limit Of Quantitation (LOQ) values of Sitagliptin incorporated in Organic impurities by HPLC method of analysis as per analytical method validation. ➤ Mobile phase stability, solution stability, typical chromatograms of standard solution, injection sequence harmonization, note point inclusion at calculation incorporated in Assay by HPLC method of analysis as per analytical method validation. ➤ Injection sequence harmonized accordingly system suitability criteria updated in residual solvents by GC method of analysis.	CCQ22069
04	26 JAN 2023	➤ Product code changed from 'SIT' to 'SP'. ➤ Nitrosamines content by LCMS (Method-I) test parameter included as per analytical method validation.	CCM22117

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 Lee Pharma Limited Unit-I, Gaddapotharam		SPECIFICATIONS AND TEST PROCEDURE	
		STP No. : STP/FP/SP/002-08	Page No. : 40 of 41
Product Name : Sitagliptin Phosphate		Department : QUALITY CONTROL	
Rev. No.	Effective Date	Details of Revision	Ref. Change Control No.
		➤ Description method of analysis updated as 'By visual observation' from 'Weighing method'. ➤ Number of sample solutions harmonized in Water content by KF.	
05	14 OCT 2023	In Nitrosamines content by LCMS, ➤ NEIPA and NDIPA impurities excluded. ➤ NTPy alternative code (NTTP) included and a separate new method included for NTPy / NTTP impurity as per analytical method validation.	CC/Q/23/210/U1
06	26 OCT 2023	➤ NTPy / NTTP Nitrosamine impurity content by LC – MS acceptance criteria revised to "Not more than 0.29 ppm" from "Not more than 0.14 ppm".	CC/Q/23/226/U1
07	09 FEB 2024	➤ NTPy / NTTP Nitrosamine impurity content by LC – MS method of analysis was updated to Not more than 0.29 level from Not more than 0.14 ppm level in line with current analytical method validation.	CC/Q/24/029/U1
08	10 / 07 / 2024	➤ Method reference numbers updated for Enantiomeric purity by HPLC, Organic impurities by HPLC, Assay by HPLC, Residual solvents by GC and Acetic acid content by HPLC test parameters. ➤ Note point included in Identification by IR test procedure.	CC/QC/24/146/U1

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 Lee Pharma Limited Unit-I, Gaddapotharam		SPECIFICATIONS AND TEST PROCEDURE	
		STP No. : STP/FP/SP/002-08	
		Department : QUALITY CONTROL	
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Product Name : Sitagliptin Phosphate			
Rev. No.	Effective Date	Details of Revision	Ref. Change Control No.
08	10 / 07 / 2024	➤ Identification by IR acceptance criteria updated to 'The Infra-red spectrum of sample should be concordant to that of Sitagliptin Phosphate CRS / Working standard' from 'The Infra-red spectrum of sample should be concordant to that of Sitagliptin Phosphate standard'. ➤ Enantiomeric purity by HPLC Identification test acceptance criteria updated to 'Compare the retention time of major peak obtained in sample chromatogram should be match with that of chromatogram of system suitability solution' from 'Compare the retention time of the major peak in the chromatogram of the sample preparation to that in the chromatogram of the standard, as obtained with system suitability solution.	CC/QC/24/146/U1

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	NEXT WAVE (INDIA) RAMPUR GHAT ROAD, PAONTA SAHIB SIRMOUR (HP) – 173025		MASTER
	RAW MATERIAL STANDARD TESTING PROCEDURE		
MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
STP No.	RM/STP/NFRD/USP/003	Revision No.	01
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1.0 Description

Examine about 1.0 g of the sample on a watch glass with white background. The sample should comply with the requirements and should be free from any foreign matter. White or almost white powder.

2.0 Solubility

Soluble in water, very slightly soluble in anhydrous ethanol and practically insoluble in heptane.

3.0 Identification

A. Infrared absorption spectrophotometry:

Take about 2 mg of test sample and mix with about 200 mg of dry potassium bromide and grind it thoroughly in mortar. Prepare a thin palate of it and collect spectra of it over previously run blank spectra of potassium bromide.

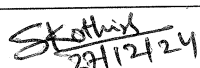
Result: The Infrared absorption spectrum of the sample should be concordant to that of Sitagliptin phosphate standard.

B. Enantiomeric Purity By HPLC:

Result: The principal peak in the chromatogram obtained with the sample solution should be correspond to the principal peak in the chromatogram obtained with system suitability solution.

C. Phosphate Test: Take about 800 mg of sample in 20 ml volumetric and add 10 ml of water and shake and sonicate to dissolve, make up the volume with water.

Result: Meets the requirements of test A of Orthophosphates.

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NAME	Jay Prakash Dubey	Saurabh Kothiya	W. Rahman
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SOP No: R&D-026

Format No- R&DF/026/01

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	NEXT WAVE (INDIA) RAMPUR GHAT ROAD, PAONTA SAHIB SIRMOUR (HP) – 173025		MASTER
RAW MATERIAL STANDARD TESTING PROCEDURE			
MATERIAL NAME		DEPARTMENT	
Sitagliptin Phosphate USP		RESEARCH & DEVELOPMENT	
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4.0 Assay

Assay determine by liquid Chromatography.

Chromatographic condition:

Column : (150x4.6)mm, 5.0 µm packing L10
Detector : UV
Wavelength : 205 nm
Flow rate : 1.0 ml/minute
Injection volumes : 20 µL
Column Temperature : 30°C

Buffer: Dissolve 1.36 gm of monobasic potassium into 1000 ml water and adjust pH to 2.0 with ortho phosphoric acid.

Mobile Phase: Mixed 85 volume of Buffer and 15 volume of Acetonitrile, filtered and degassed.

Dilute Phosphoric Acid: Dissolve 1 ml of phosphoric acid in 1000 ml water.

Diluent: Mixed 50 volumes of Acetonitrile and 950 volumes of dilute Phosphoric acid.

Standard solution: Weight accurately about 25 mg of Sitagliptin working standard and transfer in to a 25 ml of volumetric flask. Add 15 ml of diluent to dissolve and make up the volume with diluent. Further dilute 5 ml of this solution to 50 ml with diluent. (100 ppm)

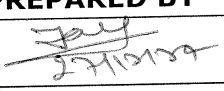
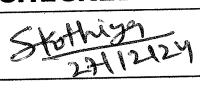
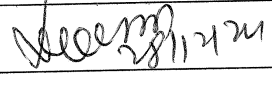
Sample Solution: Weight accurately about 50 mg of sample and transfer in to a 50 ml of volumetric flask. Add 20 ml of diluent and dissolve and make up volume with diluent and mix. Further dilute 5 ml of this solution to 50 ml with diluent. (100 ppm)

System Suitability:

Use standard solution as sample for system suitability.

System Suitability Criteria:

Relative standard deviation should NMT 0.73%

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MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
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Injection Sequence:

Sr. No.	Description	No. of Injection
1	Blank (Diluent)	01
2	Standard solution	05
3	Sample solution	02
4	Standard solution (BKT)	01

Calculation: Calculate the percentage of sitagliptin phosphate ($C_{16}H_{15}F_6N_5O \cdot H_3PO_4$) in the portion of Sitagliptin phosphate taken.

$$\frac{At}{As} \times \frac{Ws}{25} \times \frac{5}{50} \times \frac{50}{Wt} \times \frac{50}{5} \times \frac{P}{[100 - (\text{Water} + \text{Total organic solvents in per cent})]} \times 100$$

Where;

At = Area of Sample

As = Area of Standard

Ws = Weight of Sitagliptin phosphate working standard in mg

Wt = Weight of sample in mg

P = Potency of Sitagliptin phosphate working Standard

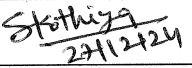
Limit: 98.0% to 102.0% on the anhydrous and solvent-free basis.

5.0 Organic Impurities

Organic Impurities determine by liquid Chromatography.

Preparation of Buffer, Mobile Phase, Dilute Phosphoric acid, Diluent and chromatographic condition are same as in Assay

System Suitability Solution: Weight accurately about 10 mg of Sitagliptin Phosphate and 1 mg of sodium stearyl fumarate into a vial, add 1 ml of water, and tightly seal the vial. Heat at

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MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
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80°C for about 30 hours to generate a fumarate adduct of sitagliptin. Transfer the content of the vial into a 100 ml volumetric flask using a small amount of diluent and make up to volume with diluent. Then mix by stirring for 1 hour after this centrifuge a portion of the solution for 10 minutes (at 1500 rpm) or until the solution is clear and use the supernatant.

Note the relative retention time for sitagliptin and fumarate adduct of sitagliptin are 1.0 and 1.2 respectively.

Preparation of Standard Solution: Weight accurately and transfer about 25 mg of Sitagliptin working standard in 250 ml volumetric flask add 70 ml of diluent and sonicate to dissolve and make up volume with diluent. Further dilute 1 ml of the above solution to 100 ml with diluent. Then further dilute 1ml of this solution to 10 ml. (0.1 ppm)

Sample solution: Weight accurately about 25 mg of sample and transfer in to a 250 ml of volumetric flask. Add 150 ml of diluent and dissolve and make up volume with diluent and mix.

Injection Sequence:

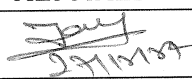
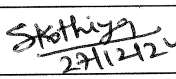
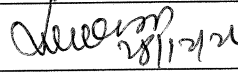
Sr. No.	Description	No. of Injection
1	Blank (Diluent)	01
2	System suitability solution	01
3	Standard solution	01
4	Sample solution	01

Equilibrate the system and conditioning the column before starting the sequence.

System Suitability Criteria:

Resolution NLT 1.5 between Sitagliptin and fumarate adduct of Sitagliptin.

Calculation: Calculate the percentage of each impurity in the portion of Sitagliptin phosphate taken.

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$$\frac{A_t}{A_s} \times \frac{W_s}{250} \times \frac{1}{100} \times \frac{1}{10} \times \frac{250}{W_t} \times \frac{P}{100} \times 100$$

Where;

A_t = Peak response of each impurity from the sample solution

A_s = Peak response of Sitagliptin from the standard solution

W_s = Weight of Sitagliptin phosphate working standard in mg

W_t = Weight of sample in mg

P = Potency of Sitagliptin phosphate working Standard

Acceptance Criteria: Disregard any peak below 0.05%.

Any individual impurity: NMT 0.1%

Total impurities: NMT 0.5%

6.0 Enantiomeric Purity

Enantiomeric Purity determine by liquid Chromatography.

Chromatographic condition:

Column : 4.6mm x 250cm; 5.0 µm packing L51

Detector : UV

Wavelength : 268 nm

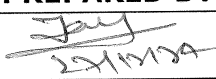
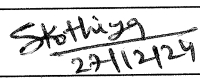
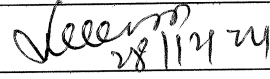
Flow rate : 0.8 ml/minute

Injection volumes : 10 µL

Column Temperature : 35°C

Mobile Phase: Mixed 600 volume of Dehydrated alcohol, 400 volume of Chromatographic n-heptane, 1 volume of diethylamine, and 1 volume of water, filtered and degassed.

Diluent: Mixed 9 volume of Methanol and 1 volume of water.

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System Suitability Solution

Weight accurately about 12 mg of Sitagliptin System Suitability Mixture reference standard into a vial, add 1.5 ml of diluent. (8000 ppm)

Sample Solution: Weight accurately about 80 mg of sample and transfer to a 10 ml of volumetric flask. Add 5 ml of diluent, dissolve and make up volume with diluent and mix. (8000 ppm)

Sensitivity Solution: Further dilute 1 ml of sample solution to 100 ml and then further dilute 1 ml of this solution to 10 ml. (8 ppm)

System Suitability:

Use System Suitability Solution and sensitivity solution as sample solution for system suitability.

Note: The relative retention times for sitagliptin, which is the R-enantiomer, and the S-enantiomer are 1.0 and 0.9, respectively.

System Suitability Acceptance Criteria:

Resolution: NLT 1.5 between the S-enantiomer and sitagliptin in system suitability solution.

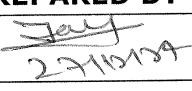
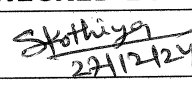
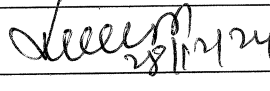
Signal to noise ratio: NLT 10 for sitagliptin peak in sensitivity solution.

Injection Sequence:

Sr. No.	Description	No. of Injection
1	Blank (Diluent)	01
2	System suitability solution	01
3	Sensitivity solution	01
4	Sample solution	01

Calculation:

$$\frac{R_u}{R_s} \times 100$$

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	RAW MATERIAL STANDARD TESTING PROCEDURE		
MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
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Where

R_u = Peak response of the S-enantiomer from the sample solution

R_s = Sum of the peak response of the S-enantiomer and sitagliptin from the sample solution

Limit: NMT 0.5% of the S-enantiomer

7.0 Water Content by KF

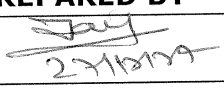
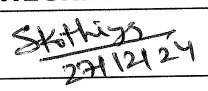
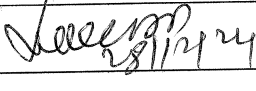
Limit: 3.3% to 3.7%

Add about 35 ml of dehydrated methanol into the titration vessel and titrate to the electrometric end point with K_F reagent. Transfer quickly about 0.300 g of the sample to the titration vessel, stir for 1 minute and titrate again to the electrometric end point by using K_F reagent.

8.0 Residue on Ignition

Analysis: Ignite a suitable crucible at $600 \pm 50^\circ\text{C}$ for 30 minutes, cool the crucible in a desiccator and weigh it accurately (W_1), weight accurately about 1 gm of substances in the Crucible (W_2).

Moisten the sample with a small amount (usually 1 ml) of sulfuric acid, then heat gently at a temperature as low as practicable until the sample is thoroughly charred. Cool; then, moisten the residue with a small amount (usually 1 ml) of sulfuric acid; heat gently until white fumes are no longer evolved; and ignite at $600 \pm 50^\circ$ for 3 hours, until the residue is completely incinerated. Ensure that flames are not produced at any time during the procedure. Cool the crucible in a desiccator (silica gel or other suitable desiccant), weigh accurately (W_3), and calculate the percentage of residue. if the amount of the residue so obtained exceeds the limit specified in the individual monograph, repeat the moistening with sulfuric acid, heating and igniting as before, using a 30-minute ignition period, until two consecutive weighing of the residue do not differ by more than 0.5 mg.

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SIGNATURE/ DATE	 23/12/24	 23/12/24	 23/12/24
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NEXT WAVE (INDIA)

RAMPUR GHAT ROAD, PAONTA SAHIB
SIRMOUR (HP) – 173025

MASTER

RAW MATERIAL STANDARD TESTING PROCEDURE

MATERIAL NAME		DEPARTMENT	
Sitagliptin Phosphate USP		RESEARCH & DEVELOPMENT	
STP No.	RM/STP/NFRD/USP/003	Revision No.	01
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$$\text{Residue of ignition \%} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \times 100$$

Where;

W_1 = Weight of empty crucible.

W_2 = Weight of crucible with test substance before ignition.

W_3 = Weight of crucible with test substance after ignition.

Limit: Not more than 0.20% w/w

9.0 *Residual solvents

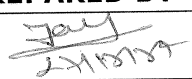
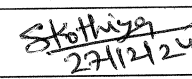
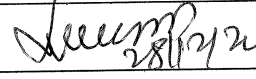
Residual solvent determine by gas Chromatography.

Chromatographic condition:

Column	DB-624 (Length = 60 met, ID = 0.53 mm, Film Thickness = 3.0µm) or equivalent
Initial Temperature	40°C
Initial hold	15 minutes
Heating Rate	20°C /min
Final Temperature	220°C
Final hold	15 minutes
Carrier gas	Nitrogen
Carrier Flow	2.5 ml/min
Spilt Ratio	1:4
Run time	39 minutes

Detector Parameter:

Detector	Flam ionization detector
Detector Temperature	250°C
Makeup flow	35.0 ml/min
Hydrogen flow	40.0 ml/min
Air flow	400 ml/min

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SIGNATURE/ DATE			
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SIRMOUR (HP) – 173025

MASTER

RAW MATERIAL STANDARD TESTING PROCEDURE

MATERIAL NAME		DEPARTMENT	
Sitagliptin Phosphate USP		RESEARCH & DEVELOPMENT	
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Head Space Sampler (HSS) conditions:

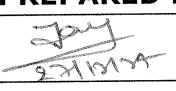
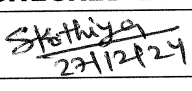
Oven temperature	90°C
Sample line Temperature	110°C
Transfer Line Temperature	110°C
Equilibrating time	30 min
Pressurize Time	2.0 min
Thermo state Time	10 min
Load Time (Loop fill)	0.50 min
Load Equilibration Time	0.10 min
Injection time	0.5 min
Injection volume	0.1 ml
Multi injection count	1
Needle flush time	5.0 min
GC cycle time	60.0 min
Pressurize gas pressure	Should be 80.0 kpa

Diluent: Use N,N-Dimethyl acetamide as a diluent.

Blank Preparation: Take 2.0 ml of diluent and 1.0 ml of HPLC grade water, transfer into a head space vial. Place the septa and immediately crimp the vial.

Chloroform Standard stock solution: Weight accurately and transfer about 60 mg of Chloroform Standard in to 50 ml volumetric flask containing about 20 ml diluent, mix well and dilute to volume with diluent.

Standard Stock solution: Weight accurately and transfer about 300 mg of Methanol Standard, 500 mg of Acetone Standard, 500 mg of Iso propyl alcohol Standard, 41 mg of Acetonitrile Standard, 60 mg of Dichloromethane Standard, 500 mg of Methyl tertiary butyl

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	NEXT WAVE (INDIA) RAMPUR GHAT ROAD, PAONTA SAHIB SIRMOUR (HP) – 173025		MASTER
	RAW MATERIAL STANDARD TESTING PROCEDURE		
MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
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ether Standard, 500 mg n-Butanol Standard, 89 mg of Toluene Standard and 5 ml of Chloroform Standard stock solution in to 50 ml volumetric flask containing about 20 ml diluent, mix well and dilute to volume with diluent.

Standard solution: Transfer 5 ml of Standard stock solution in to 100 ml volumetric flask containing about 40 ml diluent, mix well and dilute to volume with diluent. Transfer 2 ml of Standard solution and 1 ml HPLC grade water in to 20 Head space vial and immediately crimp the vial.

Sample solution: Weight accurately about 200 mg of Sample to be examined, transfer in to 20 ml head space vial add 2 ml of diluent and 1.0 ml of then immediately crimp the vial.

Injection Sequence:

Sr. No.	Description	No. of Injection
1	Blank	01
2	Standard solution	06
3	Blank	01
4	Sample solution	01

System Suitability:

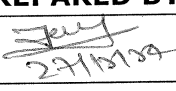
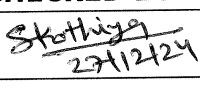
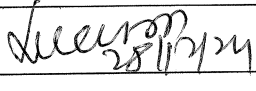
Use standard solution as sample for system suitability.

Resolution: the resolution between any two closest solvent peaks should be not less than 2.0.

% RSD for the peak area of all known solvents obtained from six injections should be not more than 15.0%

Calculation:

Calculate the ppm content of all known solvents (except Chloroform) from sample solution by using the following formula:

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$$\frac{A_t}{A_s} \times \frac{W_s}{50} \times \frac{5}{100} \times \frac{2}{3} \times \frac{3}{W_t} \times \frac{P}{100} \times 10^6$$

Where;

A_t = Peak area of respected solvent from sample Chromatogram

A_s = Average Peak area of respected solvent from six injections of standard Chromatogram

W_s = Weight of respected solvent standard in mg

W_t = Weight of sample in mg

P = Purity of respected solvent

Calculate the ppm content of Chloroform from sample solution by using the following formula:

$$\frac{A_t}{A_s} \times \frac{W_s}{50} \times \frac{5}{50} \times \frac{5}{100} \times \frac{2}{3} \times \frac{3}{W_t} \times \frac{P}{100} \times 10^6$$

Where;

A_t = Peak area of Chloroform from sample Chromatogram

A_s = Average Peak area of Chloroform from six injections of standard Chromatogram

W_s = Weight of Chloroform standard in mg

W_t = Weight of sample in mg

P = Purity of Chloroform

Limit:

Methanol NMT 3000 ppm

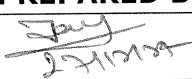
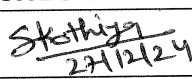
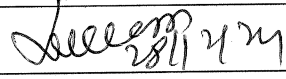
Acetone NMT 5000 ppm

Iso Propyl alcohol NMT 5000 ppm

Acetonitrile NMT 410 ppm

Dichloromethane NMT 600 ppm

Methyl tertiary butyl ether NMT 5000 ppm

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n-Butanol : NMT 5000 ppm
Toluene : NMT 890 ppm
Chloroform : NMT 60 ppm

10.0 *Acetic acid content

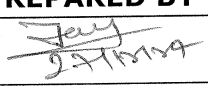
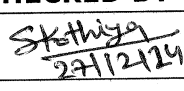
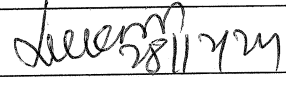
Acetic acid content determine by liquid Chromatography.

Chromatographic condition:

Column : Inertsil ODS-3 (250 x4.6) mm, 5.0 µm
Detector : UV
Wavelength : 210 nm
Flow rate : 1.0 ml/minute
Injection volumes : 10 µL
Column oven Temperature : 30°C
Auto sample Temperature : 5°C
Mode : Gradient

Gradient Program:

Time (minute)	% Mobile phase A	% Mobile phase B
0.01	100	0
10.00	100	0
13.00	10	90
29.00	10	90
30.00	100	0
45.00	100	0
45.01	Controller	Stope

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	RAW MATERIAL STANDARD TESTING PROCEDURE		
MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
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Mobile Phase A: Transfer 1 ml orthophosphoric acid into beaker containing 1000 ml of HPLC grade water, mix well. Filter through 0.45μ membrane filter paper.

Mobile Phase B: Acetonitrile.

Diluent: Methanol.

Standard solution: Weight accurately about 50 mg of Acetic acid working standard and transfer in to a 100 ml of volumetric flask. Add 40 ml of diluent to dissolve and make up the volume with diluent. Further dilute 5 ml of this solution to 50 ml with diluent. (50 ppm)

Sample Solution: Weight accurately about 100 mg of sample and transfer in to a 10 ml of volumetric flask. Add 5 ml of diluent and dissolve and make up volume with diluent and mix.

System Suitability:

Use standard solution as sample for system suitability.

System Suitability Criteria:

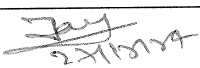
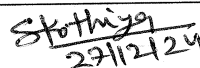
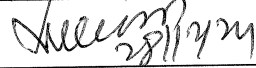
Relative standard deviation should NMT 5.0%

Injection Sequence:

Sr. No.	Description	No. of Injection
1	Blank (Diluent)	01
2	Standard solution	06
3	Sample solution	01
4	Standard solution (BKT)	01

Calculate the ppm content of Acetic acid from sample solution by using the following formula:

$$\frac{At}{As} \times \frac{Ws}{100} \times \frac{5}{50} \times \frac{10}{Wt} \times \frac{P}{100} \times 10^6$$

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Where;

At = Peak area of Acetic acid from sample Chromatogram

As = Average Peak area of Acetic acid from six injections of standard Chromatogram

Ws = Weight of Acetic acid standard in mg

Wt = Weight of sample in mg

P = Purity of Acetic acid

Limit: NMT 5000 ppm

11.0 *Phosphate content

Phosphate content determine by Potentiometrically.

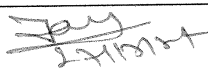
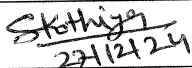
Blank titration: Transfer 50.0 ml of purified water in to 100 ml beaker, then dip the electrode in to a beaker and titrate against 0.1N sodium hydroxide solution, determine the end point potentiometrically.

Sample preparation and titration: Weight accurately and transfer about 0.3 gm of sample to be examined into 100 ml beaker, add 50 ml of purified water, then stir about 1.0 minute until dissolve the sample. Then dip the electrode into a beaker and titrate against 0.1N sodium hydroxide solution, determine the end point potentiometrically.

Calculate the Phosphate Content (On Anhydrous basis) in sample solution by using below formula.

$$\% \text{ Phosphate Content} = \frac{\text{Titer volume} \times \text{Normality of 0.1N sodium hydroxide} \times 49 \times 100}{0.1 \times \text{Weight of sample} \times 1000} \times 100$$

$$\% \text{ Phosphate Content (On Anhydrous basis)} = \frac{\% \text{ Phosphate Content}}{(100 - \text{water content of sample})} \times 100$$

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Where,

Titer volume = Sample titer volume – Blank titer volume

The titer value for blank should be not more than 0.1 ml.

12.0 *NTPy / NTTP Nitrosamine impurity content

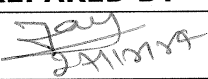
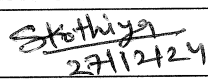
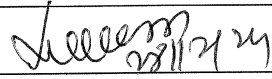
NTPy / NTTP Nitrosamine impurity content determine by LC-MS.

Chromatographic condition:

Instrument	: LC-MS
Column	: Shimadzu, shim pack GIST C18 (250 x4.6) mm, 5.0 µm
Detector	: UV
Flow rate	: 0.7 ml/minute
Injection volumes	: 20 µL
Column oven Temperature	: 45°C
Auto sample Temperature	: 15°C
Run Time	: 28 minutes
Mode	: Gradient

Gradient Program:

Time (minute)	% Mobile phase A	% Mobile phase B
0.01	80	20
2.00	80	20
7.00	60	40
12.00	45	55
15.00	05	95
19.00	05	95
20.00	80	20

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Mobile Phase A: Transfer 1 ml formic acid into beaker containing 1000 ml of LCMS grade water, mix well and sonicate to degas. Then filter through 0.45μ membrane filter paper.

Mobile Phase B: Transfer 1 ml formic acid into beaker containing 1000 ml of LCMS grade methanol, mix well and sonicate to degas. Then filter through 0.45μ membrane filter paper.

Diluent: Ethyl acetate use as diluent.

Needle wash solution: Methanol should be use as needle wash solution.

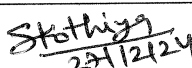
Blank solution: Diluent use as blank solution.

Interface parameters:

Interface : ESI
Acquisition Mode : MRM
Polarity : Positive
Nibulizing Gas Flow : 2.00 L/minute
Interface Temperature : 200°C
DL Temperature : 150°C
Heat block Temperature : 200°C
Drying gas flow : 10.0 L/minute
Heat gas flow : 10.0 L/minute

MRM Transitions:

Name	Precursor m/z	Product m/z	Q1 Pre Bias (V)	Collision Energy	Q3 Pre Bias (V)
NTPy / NTPP	222.10	192.15	-16	-10	-20

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MS program Valve Position:

Initial valve position:1

Time	Command	Valve
0.10	FCV2=	1
10.00	FCV2=	0
25.00	FCV2=	1

NTPy / NTPP Standard stock solution: Weigh accurately and transfer about 9.2 mg of NTPy/NTPP standard in to 15 ml of tarson tube, then add 5.0 ml of methanol. Mix well to dissolve and make up with methanol up to 10 ml mark.

Standard stock solution-1: Transfer accurately 0.8 ml of NTPy/NTPP standard stock solution into 50 ml of tarson tube containing about 20 ml of methanol. Mix well and make up to mark with methanol.

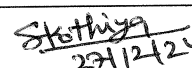
Standard stock solution-2: Transfer accurately 0.5 ml of standard stock solution-1 into 50 ml of tarson tube containing about 20 ml of diluent. Mix well and make up to mark with diluent.

Standard solution: Transfer accurately 1.0 ml of standard stock solution-2 into 50 ml of tarson tube containing about 20 ml of diluent. Mix well and make up to mark with diluent.

Solution stability: Standard solution was stable up to 29 Hours at room temperature (25°C) and at 38°C.

Sensitivity solution: Transfer accurately 0.10 ml of standard stock solution-2 and transfer into 50 ml of tarson tube containing about 20 ml of diluent. Mix well and make up to mark with diluent.

Sample solution: Weigh accurately about 100.0 mg of sample to be examined and transfer in to 15 ml centrifuged tube, add 10 ml of diluent and vortex for 60 seconds. Then sonicate for 5 minutes and mix well on vortex mixture for about 60 seconds. Then centrifuge the

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sample solution for 10 minutes at 4500 rpm speed. Filter the sample solution through 0.22 µm nylon syringe filters.

Solution stability: Sample solution was stable up to 25 Hours at room temperature (25°C) and at 35°C.

Note: The retention time of NTPy / NTTP peak is about 8.72 minutes

System suitability:

Use sensitivity solution and Standard solution as sample for system suitability.

System suitability criteria:

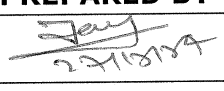
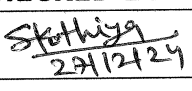
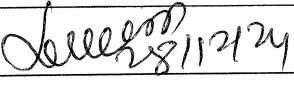
The S/N Ratio for the NTPy / NTTP peak is should be not less than 10.0, Sensitivity solution Relative standard deviation for the peak areas of NTPy / NTTP obtained from six replicate injections is should be not more than 20.0%, Standard solution

Injection Sequence:

Sr. No.	Description	No. of Injection
1	Blank solution	01 or 02
2	Sensitivity solution	01
3	Standard solution	01
4	Blank solution	01
5	Standard solution	06
6	Blank solution	01
7	Sample solution	01

Calculate the ppm content of NTPy / NTTP from sample solution by using the following formula:

$$\frac{At}{As} \times \frac{Ws}{10} \times \frac{0.8}{50} \times \frac{0.5}{50} \times \frac{1.0}{50} \times \frac{10}{Wt} \times \frac{P}{100} \times 10^6$$

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Where;

At = Peak area of NTPy / NTTP from sample Chromatogram

As = Average Peak area of NTPy / NTTP from six injections of standard Chromatogram

Ws = Weight of NTPy / NTTP standard in mg

Wt = Weight of sample in mg

P = Purity of NTPy / NTTP impurity standard

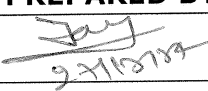
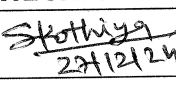
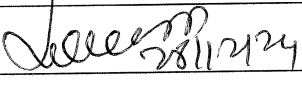
Correct peak area = (Peak area of area respective impurity from standard/sample) - (Peak area of area respective impurity from blank)

Retention Time (RT), Limit of Detection (LOD) and Limit of Quantitation (LOQ) values of NTPy given in below table;

Name	LOD conc. In ppm w.r.t. sample	LOQ conc. In ppm w.r.t. sample
NTPy / NTTP	0.0120	0.0363

Limit: NMT 0.29 ppm

***Additional Test**

	PREPARED BY	CHECKED BY	APPROVED BY
SIGNATURE/ DATE			
NAME	Jay Prakash Dubey	Saurabh Kothiya	W. Rahman
DESIGNATION	Sr. Officer	Executive	Dy. Manager QA

SOP No: R&D-026

Format No- R&DF/026/01

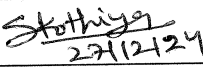
UNCONTROLLED

	NEXT WAVE (INDIA) RAMPUR GHAT ROAD, PAONTA SAHIB SIRMOUR (HP) – 173025		MASTER
	RAW MATERIAL STANDARD TESTING PROCEDURE		
MATERIAL NAME Sitagliptin Phosphate USP		DEPARTMENT RESEARCH & DEVELOPMENT	
STP No.	RM/STP/NFRD/USP/003	Revision No.	01
Effective Date	30/12/2024	Supersedes	00
		Page No.	Page 20 of 20

13.0 Revision History

Revision No.	Supersedes	Effective Date	Details of Revision
00	Nil	08/01/2022	New STP
01	00	30/12/2024	Incorporate test of Residue on Ignition, Phosphate content by Potentiometric & NTPy / NTTP Nitrosamine impurity content CC/AL/24/091

End of Document

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SIGNATURE/ DATE			
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DESIGNATION	Sr. Officer	Executive	Dy. Manager QA

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